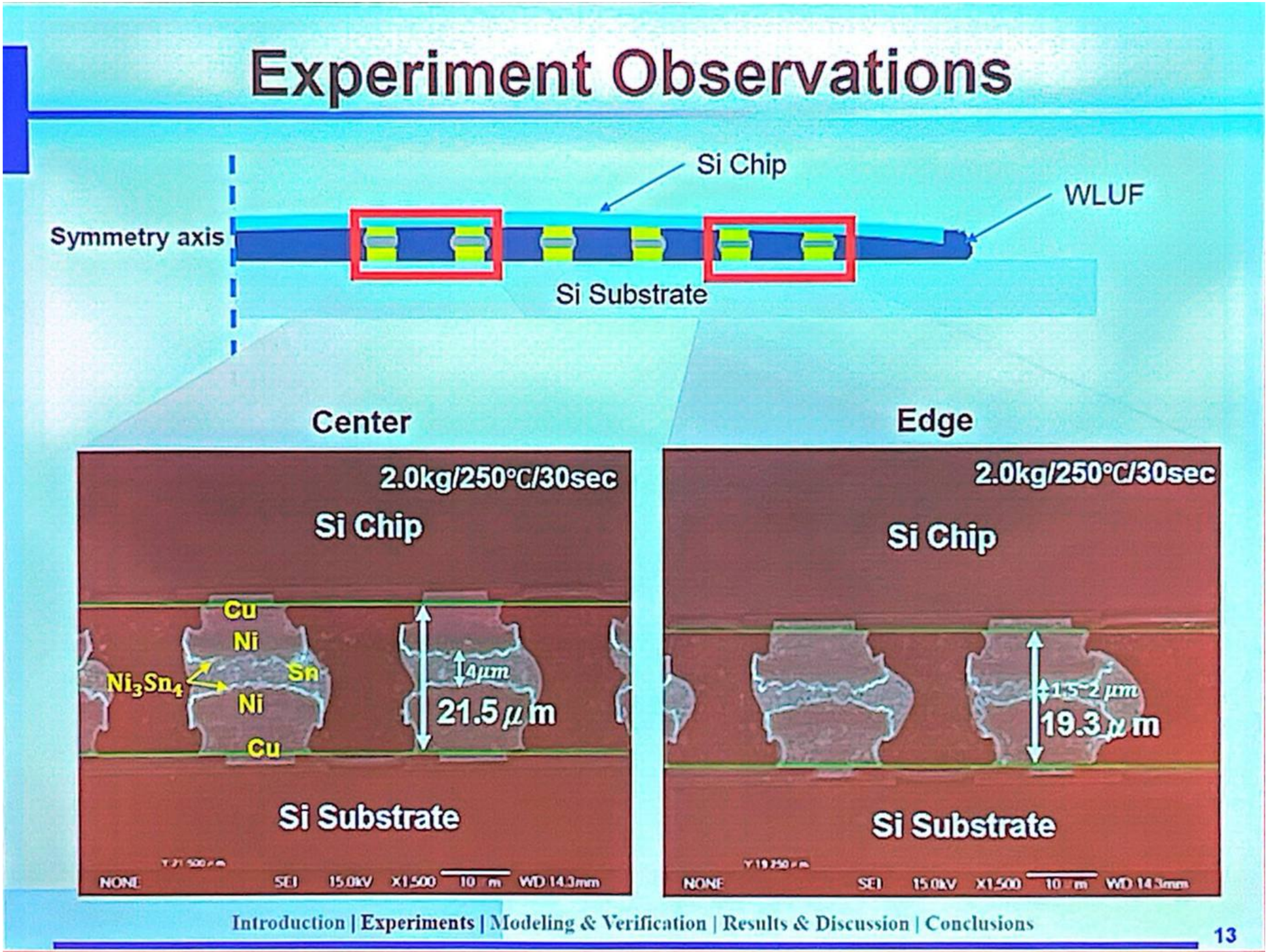
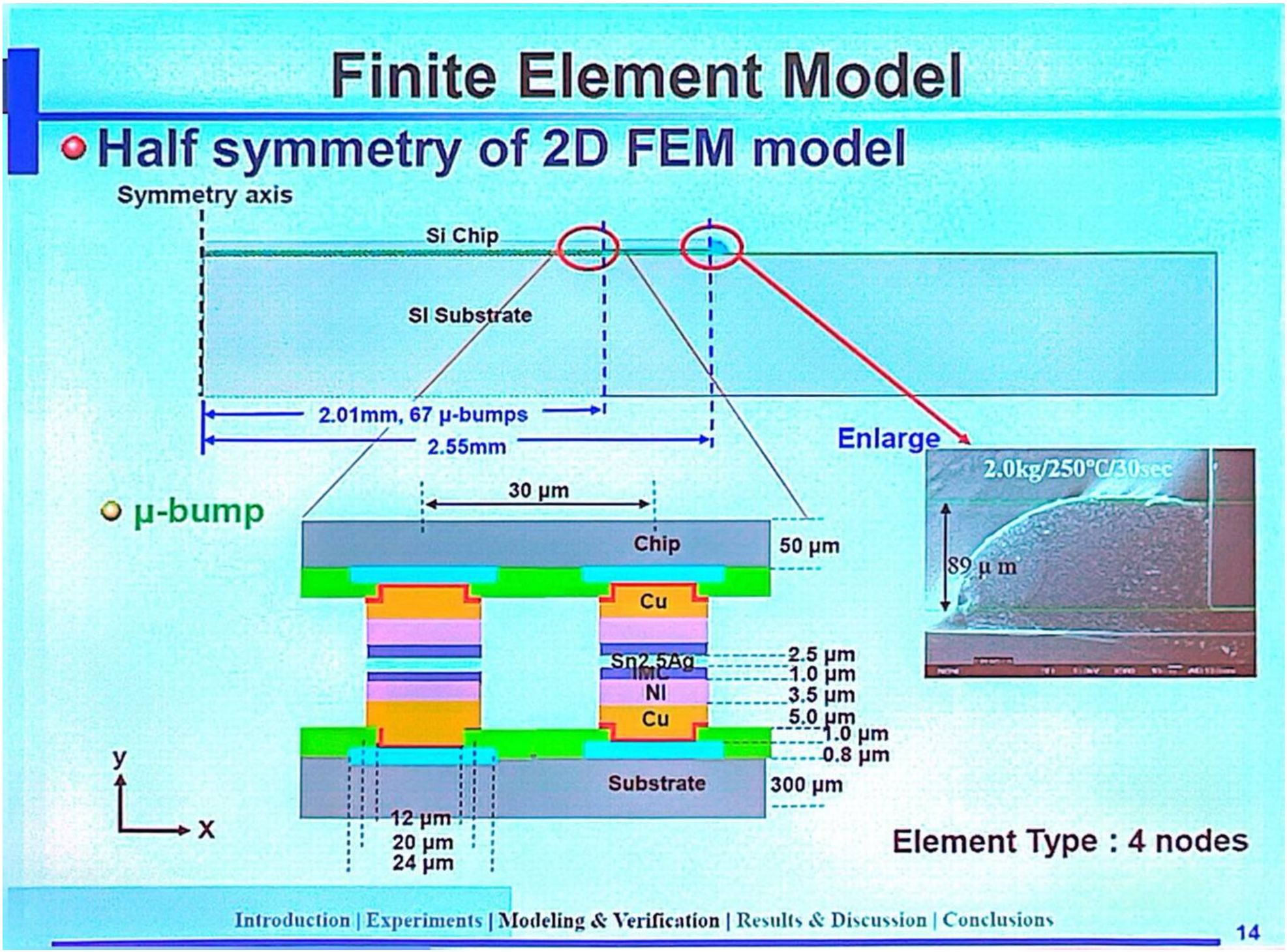
T33 WLUF 기반 3D 적층의 하중과 휨

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Material Properties

Materials	Elastic Modulus (E)	CTE (ppm/K)		Poisson's Ratio
Si	169.5 GPa	3		0.28
Cu (bilinear Model)	E = 122 GPa Ref. T = 25°C σ_Y = 173 MPa E_T = 1.2 GPa	17		0.35
Passivation (Si_3N_4)	155 GPa	5		0.28
Wafer-level Underfill	6.8 GPa @ 25°C	53		0.33
SnAg solder	Viscoplastic Model	22.5		0.4
IMC	85.6 GPa	-40°C	16.3	0.31
		25°C	17.6	
		50°C	18.1	
		125°C	19.3	
Ni	186 GPa	12.5		0.342
Al	72 GPa	24		0.36

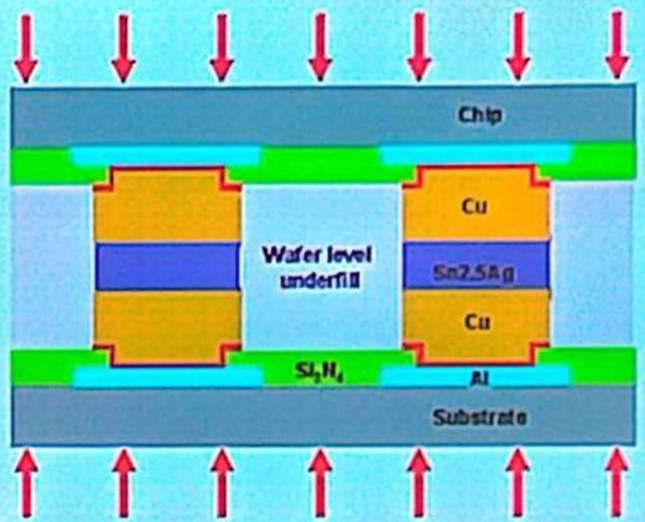
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Process Oriented Simulation

STEP1- Apply the external pressure (bonding force) at T= 250°C.

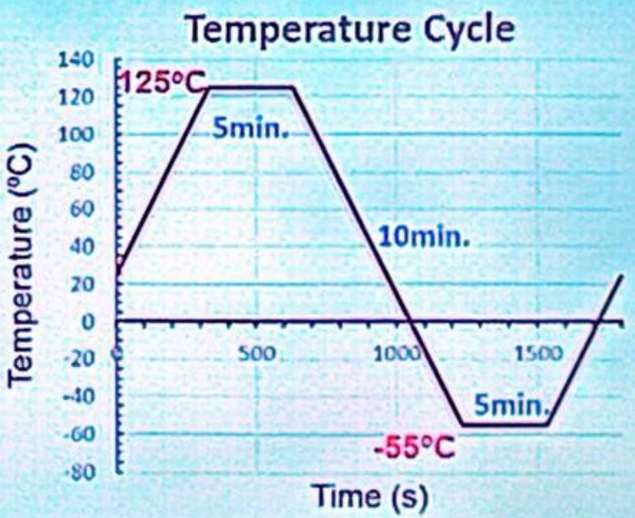
2.0kg/250°C/30sec

The Wafer level underfill (WLUF) is in liquidity status for FEA, and the bumps are in bonding condition.



STEP2- Curing the WLUF at T= 250°C.

STEP3- Release the bonding force at T= 250°C.



STEP4- Cool down to room temperature of 25°C.

STEP5- Apply thermal loading under JEDEC JESD22-A-104 D

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Conclusions

- **Key Findings**
  - **High Reliability:**

    Characteristic lifetime exceeding 1250 cycles under -55°C ~ 125°C thermal loads

  - **Promising 2D Finite Element Model:**

    The model predicted a 1.84μm warpage at the chip edge, which closely matched the experimental measurement of approximately 2μm

  - **Identification of Failure Mechanisms:**

    Simulation and SEM shows plastic strain concentration and cracks in the lead-free solder region adjacent to the brittle Ni_3Sn_4 IMC layer